

```
>> three_dof_robot
```

```
solve_thetal56_left =
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```
[ nx*cos(theta1)*cos(theta6) + ny*cos(theta6)*sin(theta1) - ox*cos(theta1)*sin(theta6) - oy*cos(theta1)*sin(theta6), ox*cos(theta1)*cos(theta6) + nx*cos(theta1)*sin(theta6) + oy*cos(theta6)*sin(theta1) + ny*sin(theta1)*sin(theta6), ax*cos(theta1) + ay*sin(theta1), px*cos(theta1) + py*sin(theta1) - ay*d6*sin(theta1) - ax*d6*cos(theta1)]
[
                                oz*sin(theta6)
- nz*cos(theta6),
oz*cos(theta6) - nz*sin(theta6), -ax,
d1 - pz + az*d6]
[ ny*cos(theta1)*cos(theta6) - nx*cos(theta6)*sin(theta1) - oy*cos(theta1)*sin(theta6) + ox*sin(theta1)*sin(theta6), oy*cos(theta1)*cos(theta6) + ny*cos(theta1)*sin(theta6) - ox*cos(theta6)*sin(theta1) - nx*sin(theta1)*sin(theta6), ay*cos(theta1) - ax*sin(theta1), py*cos(theta1) - px*sin(theta1) + ax*d6*sin(theta1) - ay*d6*cos(theta1)]
[
0,
0,
1]
```

```
solve_thetal56_right =
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```
[ cos(theta2 + theta3 + theta4)*cos(theta5), -sin(theta2 + theta3 + theta4), -cos(theta2 + theta3 + theta4)*sin(theta5), a3*cos(theta2 + theta3) + a2*cos(theta2) + d5*sin(theta2 + theta3 + theta4)]
[ sin(theta2 + theta3 + theta4)*cos(theta5), cos(theta2 + theta3 + theta4), -sin(theta2 + theta3 + theta4)*sin(theta5), a3*sin(theta2 + theta3) + a2*sin(theta2) - d5*cos(theta2 + theta3 + theta4)]
[
                                sin(theta5),
cos(theta5),
[
                                0,
0,
1]
```

```
solve_theta234_left =
```

```
[ nx*cos(theta1)*cos(theta5)*cos(theta6) - ay*sin(theta1)*sin(theta5) - ax*cos(theta1)*sin(theta5) + ny*cos(theta5)*cos(theta6)*sin(theta1) - ox*cos(theta1)*cos(theta5)*sin(theta6) - oy*cos(theta5)*sin(theta1)*sin(theta6), ax*cos(theta1)*cos(theta5) + ay*cos(theta5)*sin(theta1) + nx*cos(theta1)*cos(theta6)*sin(theta5) + ny*cos(theta6)*sin(theta1)*sin(theta5) - ox*cos(theta1)*sin(theta5)*sin(theta6) - oy*sin(theta1)*sin(theta5)*sin(theta6), - ox*cos(theta1)*cos(theta6) - nx*cos(theta1)*sin(theta6) - oy*cos(theta6)*sin(theta1) - ny*sin(theta1)*sin(theta6), px*cos(theta1) + py*sin(theta1) - ay*d6*sin(theta1) - ax*d6*cos(theta1) + d5*ny*sin(theta1)*sin(theta6) + d5*ox*cos(theta1)*cos(theta6) + d5*nx*cos(theta1)*sin(theta6) + d5*oy*cos(theta6)*sin(theta1)]
[
az*sin(theta5) - nz*cos(theta5)*cos(theta6) + oz*cos(theta5)*sin(theta6),
oz*sin(theta5)*sin(theta6) - nz*cos(theta6)*sin(theta5) - az*cos(theta5),
oz*cos(theta6) + nz*sin(theta6)]
```

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d1 = pz + az*d6 - d5*oz*cos(theta6) - d5*nz*sin(theta6)]
[ ax*sin(theta1)*sin(theta5) - ay*cos(theta1)*sin(theta5) + ny*cos(theta1)*cos(theta5)*cos(theta6)
- nx*cos(theta5)*cos(theta6)*sin(theta1) - oy*cos(theta1)*cos(theta5)*sin(theta6) + ox*cos(theta5)
*sin(theta1)*sin(theta6), ay*cos(theta1)*cos(theta5) - ax*cos(theta5)*sin(theta1) + ny*cos(theta1)
*cos(theta6)*sin(theta5) - nx*cos(theta6)*sin(theta1)*sin(theta5) - oy*cos(theta1)*sin(theta5)*sin
(theta6) + ox*sin(theta1)*sin(theta5)*sin(theta6), ox*cos(theta6)*sin(theta1) - ny*cos(theta1)
*sin(theta6) - oy*cos(theta1)*cos(theta6) + nx*sin(theta1)*sin(theta6), py*cos(theta1) - px*sin
(theta1) + ax*d6*sin(theta1) - ay*d6*cos(theta1) - d5*nz*sin(theta1)*sin(theta6) + d5*oy*cos
(theta1)*cos(theta6) + d5*ny*cos(theta1)*sin(theta6) - d5*ox*cos(theta6)*sin(theta1)]
[
0,
0,
0,
1]

```

```

solve_theta234_right =

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```

[ cos(theta2 + theta3 + theta4), 0, sin(theta2 + theta3 + theta4), a3*cos(theta2 + theta3) +
a2*cos(theta2)]
[ sin(theta2 + theta3 + theta4), 0, -cos(theta2 + theta3 + theta4), a3*sin(theta2 + theta3) +
a2*sin(theta2)]
[
0, 1,
d4]
[
0, 0,
1]

>>

```